

Program : Diploma in Wood and Paper Technology	
Course Code : 4127	Course Title: Material Testing Lab
Semester : 4	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0, T:0, P:3)	Periods per semester: 45

Course Objectives:

- To provide knowledge on Mechanical behaviour of materials
- To acquaint with the experimental methods to determine the mechanical properties of wood

Course Prerequisites:

Topic	Course code	Course name	Semester
Material strength		Applied physics I	1
		Engineering mechanics	2

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Conduct physical and mechanical testson materials/ wood	12	Applying
CO2	Conduct hardness tests on materials/ wood	9	Applying
CO3	Conduct Impact test on materials /wood	9	Applying
CO4	Determine the mechanical properties of wood/material	12	Applying
	Lab Exam	3	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1		3		3			
CO2		3		3			
CO3		3		3			
CO4		3		3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Conduct physical and mechanical testson materials/ wood		
M1.01	Determine the tensile and shear strength of timber	3	Applying
M1.02	Conduct bending test on wooden beams and plot a graph 'load' against 'deflection' and from the graph find the young's modulus	3	Applying
M1.03	Conduct compression test on wood and find the permissible compressive stress of wood	3	Applying
M1.04	Determine the glue shear strength of plywood	3	Applying
CO2	Conduct hardness tests on materials/ wood		
M2.01	Determine thebrinell hardness by conducting brinell hardness test on M.S, Brass, wood and Aluminium.	6	Applying
M2.02	Determine the rock well hardness by conductingrock well hardness test on M.S., Brass, and Aluminium	3	Applying
	Lab Exam – I	1	
CO3	Conduct Impact test on materials /wood		
M3.01	Conduct Izod and Charpy impact tests on respective standard specimens of steel and find the energy required to break the same	6	Applying
M3.01	Determine the impact values of wood and wood laminates by impact test - izode impact test	3	Applying
CO4	Determine the mechanical properties of wood/material		
M4.01	Determine modulus of rigidity of material from angle of twist and torque byTorsion test on wood and M.S. bar	6	Applying
M4.02	Determine modulus of rupture of wood Conduct water absorbancy test on wood	6	Applying
	Lab Exam – II	2	

Text / Reference:

T/R	Book Title/Author
T1	G E Dieter. Mechanical Metallurgy, McGraw Hill, 2013
R2	Dally J W, Railey W P, Experimental Stress analysis, McGraw Hill, 1991
R3	Mechanical Behavior of Materials, by N Dowling, Prentice Hall, 1993.

Online Resources:

Sl.No	Website Link
1	https://nptel.ac.in/courses/112107147/
2	https://nptel.ac.in/courses/112103108/